EXERCISE SHEET
Properties of Expectation and Variance

EXERCISE 1 */**

Let X_1 and X_2 be two independent random variables with same mean equal to 1 and same variance equal to 4. Then

A $\mathbb{E}[(X_1 + X_2)^2] = 12$

B $\mathbb{E}[(X_1 + X_2)^2] = 10$

C $\mathbb{E}[(X_1 + X_2)^2] = 4$

D $\mathbb{E}[(X_1 + X_2)^2] = 8$

X_1 and X_2 are two independent random variables that take on the values 0 or 2, both with probability 1/2. Then

A $\mathbb{E}[\min\{X_1, X_2\}] = 1/2$

B $\mathbb{E}[X_1(X_2 + 1)] = 1$

C $\mathbb{E}[\min\{X_1, X_2\}] = 0$

D $\mathbb{E}[X_1(X_2 + 1)] = 3$

X is an exponential random variable of parameter 2 and Y a Poisson random variable of parameter 2. Suppose X and Y are independent. Then

A $\mathbb{E}[4X^2 + 3Y^2] = 20$

B $Var(4X + 3Y - 1) = 6$

C $\mathbb{E}[4X^2 + 3Y^2] = 13$

D $Var(4X + 3Y - 1) = 21$

EXERCISE 2 **

In an urn there are 8 dice. The half of them are fair, while the other half are biased, in such a way that each biased die lands showing face 2 with probability 1/2 and the other faces with equal probability. A die is chosen at random from the urn and rolled. Let X be the face shown. Compute: the probability mass function $P_X(x)$ of X, $E[X]$ and $Var(X)$.

EXERCISE 3 **

Consider the i.i.d. r.v.'s $X_1, \ldots, X_n$ such that $E[X_i] = \mu$ and $Var(X_i) = \sigma^2$. Let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

(i) Calculate $E[\bar{X}]$ and $Var(\bar{X})$

(ii) Calculate $E[\bar{X} + X_1]$ and $Var(\bar{X} + X_1)$.

EXERCISE 4 */**

X and Y are two independent binomial random variables of parameters $n = 3$, $p = 0.2$ and $n = 7$, $p = 0.2$, respectively. Then

A $\mathbb{P}[X + Y = 3] = \binom{10}{3}(0.2)^3$

B $\mathbb{P}[X + Y = 3] = \binom{7}{3}(0.2)^3(0.8)^7$

C $\mathbb{P}[X + Y = 3] = \binom{10}{3}(0.2)^3(0.8)^7$

D $\mathbb{P}[X + Y = 3] = 1$

X and Y are two uncorrelated Bernoulli random variables with parameters $\frac{1}{3}$ and $\frac{2}{3}$, respectively.

A $\mathbb{P}[XY = 0] = \frac{1}{3}$

B $\mathbb{E}[XY] = \frac{1}{9}$

C $\mathbb{P}[XY = 1] = \frac{2}{9}$

D $\mathbb{E}[X - Y] = 0$

X is a normal random variable with mean 1 and variance 4 and Y a Bernoulli random variable of parameter $\frac{1}{3}$. Suppose X and Y are independent. Then

A $\mathbb{P}[2XY > -1] = 0.7734$

B $\mathbb{P}[2XY > 1] = 0.1996$

C $\mathbb{P}[2XY > 1] = 0.5987$

D $\mathbb{P}[2XY > -1] = 0.2578$

EXERCISE 5 */**

Let X and Y be two discrete random variables with joint probability mass function represented in the table below:

X/Y	0	1
-1	0	p
0	p	0
1	1/6	1/6

Compute:

a) p

b) The marginal probability mass functions of X and Y. Are X and Y independent?

c) The expectation of X

d) The expectation and the variance of Y

e) $P(U = 0)$, where $U := XY$.

EXERCISE 6 ***

For a positive random variable X such that $X \geq 0$, show that:

$$E[X] = \sum_{x=1}^{+\infty} P(X \geq x)$$